

NVIDIA-Certified Associate Generative AI LLMs Master Cheat Sheet

Organized quick-review glossary for exam recall, scenario mapping, and final revision.

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How to use: scan domains first, then use decision tree and scenario keywords for exam questions.

Main Sections

- Exam Domains
- Repeated Exam Patterns
- AI Subject Terms Glossary
- Python Tools and Libraries
- Exam Decision Tree
- Common Exam Traps
- Formula and Metric Mini Sheet
- Scenario Keywords

Consolidated from the provided PDFs and domain screenshots. Definitions are intentionally short for fast exam review.

Exam Domains

Core ML & AI Fundamentals

Weight: 30%

Concepts

Artificial Intelligence (AI): Systems that perform tasks requiring human intelligence.

Machine Learning (ML): Algorithms learn patterns from data.

Deep Learning (DL): Multi-layer neural networks learn representations automatically.

Generative AI: AI that creates text, images, code, audio, or data.

Foundation Model: Large pretrained model adaptable to many tasks.

Large Language Model (LLM): Transformer model trained on large text corpora.

Supervised Learning: Learns from labeled input-output examples.

Unsupervised Learning: Finds patterns in unlabeled data.

Semi-supervised Learning: Uses small labeled data plus large unlabeled data.

Self-supervised Learning: Creates labels from raw data itself.

Reinforcement Learning: Learns actions using rewards.

Transfer Learning: Reuses pretrained knowledge for a new task.

Feature Extraction: Freeze base model; train only final layers.

Fine-tuning: Continue training a pretrained model on task data.

Feature Engineering: Create useful model inputs from raw data.

Representation Learning: Model learns useful internal features.

Embedding: Dense vector representing meaning or features.

Text Embedding: Vector representation of text semantics.

Word2Vec: Learns word embeddings from text context.

CBOW: Predicts target word from surrounding context.

Skip-gram: Predicts context words from a target word.

Overfitting: Performs well on training data, poorly on unseen data.

Underfitting: Model too simple to learn patterns.

Regularization: Reduces overfitting.

Dropout: Randomly disables neurons during training.

L1 Regularization: Encourages sparse weights.

L2 Regularization: Penalizes large weights.

Loss Function: Measures prediction error.

Gradient Descent: Updates weights to reduce loss.

Learning Rate: Step size for weight updates.

Backpropagation: Computes gradients through network layers.

Epoch: One full pass through training data.

Batch Size: Samples processed per training step.

Hyperparameter: Config set before training.

Parameter: Learned model weight.

Inference: Running a trained model to produce output.

Classification: Predicts categories.

Regression: Predicts continuous values.

Clustering: Groups similar unlabeled data.

Anomaly Detection: Finds unusual data points.

Association Rule Mining: Finds item co-occurrence rules.

Time Series Forecasting: Predicts future ordered values.

Data Augmentation: Creates varied training examples.

Normalization: Scales values for stable training.

Standardization: Centers/scales features by mean and variance.

Padding: Adds tokens/pixels to standardize size.

Masking: Prevents attention to irrelevant positions.

Architectures

Dense Network: Fully connected neural network.

CNN: Learns spatial features from images.

Convolution: Sliding filter extracts local patterns.

Kernel/Filter: Small matrix detecting image features.

Stride: Step size of convolution movement.

Pooling: Reduces spatial size.

Max Pooling: Keeps strongest local activation.

Average Pooling: Averages local activations.

RNN: Processes sequential data step by step.

LSTM: RNN variant for long dependencies.

GRU: Simpler gated RNN variant.

Transformer: Attention-based architecture for sequences.

Encoder-only Model: Reads input for understanding tasks.

Decoder-only Model: Generates text autoregressively.

Encoder-decoder Model: Converts input sequence to output sequence.

BERT: Bidirectional encoder model.

GPT: Decoder-only autoregressive model.

T5: Encoder-decoder text-to-text model.

BART: Encoder-decoder generation model.
Vision Transformer (ViT): Treats image patches like tokens.
Autoencoder: Compresses and reconstructs data.
VAE: Generative autoencoder with latent distribution.
GAN: Generator and discriminator compete.
Diffusion Model: Learns to denoise step by step.
CLIP: Aligns image and text embeddings.
GNN: Learns from graph nodes and edges.
Multimodal LLM: Processes multiple data types together.

Transformer Concepts

Self-attention: Tokens attend to other tokens in same sequence.
Cross-attention: Decoder attends to encoder outputs.
Multi-head Attention: Multiple attention views in parallel.
Query (Q): Vector used to search attention targets.
Key (K): Vector used for attention matching.
Value (V): Vector containing retrieved information.
Scaled Dot-product Attention: Attention using QK scores scaled by $\sqrt{d_k}$.
$\sqrt{d_k}$ Scaling: Prevents large dot products destabilizing softmax.
Positional Encoding: Adds token order information.
Sinusoidal Encoding: Original fixed transformer position signal.
RoPE: Rotary position embedding for long contexts.
Causal Mask: Prevents decoder from seeing future tokens.
Padding Mask: Ignores padded tokens.
Feed-forward Network (FFN): Per-token nonlinear transformation.
Residual Connection: Adds input back to layer output.
LayerNorm: Normalizes activations inside transformer layers.
RMSNorm: Efficient LayerNorm alternative.
Context Window: Maximum tokens a model can process.
KV Cache: Stores attention keys/values for faster decoding.
Grouped-query Attention: Shares KV heads to reduce inference memory.
Sliding Window Attention: Attends to local token windows.
FlashAttention: Faster memory-efficient attention.
Mixture of Experts (MoE): Activates only selected expert submodels.

Applications

RAG Systems: Retrieve knowledge before generation.

Chatbots: Conversational LLM applications.
Summarizers: Shorten text while preserving meaning.
Translation: Convert text between languages.
Sentiment Analysis: Classify emotional tone.
NER: Detect named entities in text.
Token Classification: Label each token.
Text Classification: Label whole text.
Question Answering: Answer from context or generation.
Extractive QA: Selects answer span from passage.
Generative QA: Produces free-form answer.
Image Classification: Predicts image class.
Object Detection: Finds objects in images.
Speech Recognition: Converts audio to text.
Fraud Detection: Flags suspicious behavior.
Customer Churn Prediction: Predicts customer loss.
Recommendation Systems: Suggest items/users/content.
Drug Discovery: Uses models for molecules/proteins.

Frameworks

PyTorch: Open-source deep learning framework.
TensorFlow: End-to-end ML platform/framework.
JAX: Autodiff and accelerated numerical framework.
Hugging Face Transformers: Pretrained transformer model library.
LangChain: Framework for LLM apps/agents.
LangGraph: Stateful graph workflows for agents.
LlamaIndex: Data/RAG framework for LLM apps.
NVIDIA NeMo Framework: Builds/customizes generative AI models.
NVIDIA NeMo Guardrails: Framework/toolkit for LLM safety rails.
RAPIDS: NVIDIA GPU data science framework/suite.

Libraries

scikit-learn: Classical ML library/toolkit.
Keras: High-level TensorFlow model API.
Hugging Face Datasets: Dataset loading/preprocessing library.
Hugging Face PEFT: Parameter-efficient fine-tuning library.
Hugging Face TRL: RLHF/DPO training utilities.
spaCy: Production NLP pipeline library.

NLTK: Traditional NLP toolkit.
SentenceTransformers: Embedding models for semantic search.
NeMo Curator: Dataset curation and filtering pipeline.
TensorRT-LLM: Optimized LLM inference library.
cuDF: GPU DataFrame library.
cuML: GPU machine learning algorithms.
cuGraph: GPU graph analytics.
Dask-cuDF: Distributed GPU DataFrames.
NVIDIA DALI: GPU data loading/preprocessing.
CuPy: NumPy-like arrays on GPU.
Numba: JIT compiler for Python/GPU kernels.
FAISS: Local vector similarity search library.
ONNX Runtime: Runs ONNX models.
NumPy: Python numeric computing package.
Pandas: CPU DataFrame analysis library.
Polars: Fast DataFrame processing library.
SciPy: Scientific computing and optimization.
Matplotlib: Basic Python plotting.
Seaborn: Statistical visualization library.
Plotly: Interactive visualization library.

NVIDIA Platforms, Microservices, and Serving Software

NVIDIA NIM: Optimized inference microservices.
Triton Inference Server: Open-source model serving software.
TensorRT-LLM: NVIDIA LLM inference optimization library.
NVIDIA AI Enterprise: Enterprise AI software platform.
NVIDIA AI Workbench: Development environment manager.
NVIDIA Base Command: AI workflow/DGX management platform.
NVIDIA AI Endpoints: Hosted NVIDIA model API service.
NGC: NVIDIA catalog for models/containers/SDKs.
BioNeMo: NVIDIA platform for biomolecular AI.
DeepStream SDK: Streaming video AI SDK.
Omniverse: 3D simulation/collaboration platform.

Vector Databases and Retrieval Platforms

Milvus: Open-source vector database.
Pinecone: Managed vector database.

Weaviate: Semantic vector database.

Chroma: Lightweight local vector database.

FAISS: Local vector similarity search library.

Development, Deployment, and MLOps Tools

ONNX: Open model exchange format.

Docker: Container runtime.

Kubernetes: Container orchestration platform.

Helm: Kubernetes package manager.

FastAPI: Modern Python REST API framework.

Flask: Lightweight Python web framework.

Streamlit: Quick Python data apps.

Gradio: Quick ML demos/interfaces.

MLflow: Experiment and model tracking.

Weights & Biases: Experiment tracking dashboard.

DVC: Data/model version control.

Ray: Distributed Python computing.

Ray Serve: Scalable model serving library.

BentoML: Model packaging/serving framework.

Airflow: Workflow orchestration.

Prefect: Python workflow orchestration.

Docker Hub: Container image registry.

GitHub: Code hosting/collaboration platform.

Jupyter Notebook: Interactive notebooks.

pytest: Python testing framework.

Great Expectations: Data quality validation.

Evidently: ML drift and monitoring reports.

Data Analysis Techniques

Weight: 14%

Techniques

Data Mining: Extracts patterns from large data.

Data Visualization: Shows insights with charts.

Statistical Metrics: Quantify model/data behavior.

Trend Analysis: Finds patterns over time.

Performance Comparison: Compares models using metrics.

Exploratory Data Analysis: Understands distributions and relationships.

Descriptive Statistics: Summarizes data.
Correlation Analysis: Measures relationship between variables.
Distribution Analysis: Studies data shape/spread.
Outlier Detection: Finds abnormal values.
Dimensionality Reduction: Compresses features while preserving structure.
PCA: Linear dimensionality reduction.
Nearest Neighbor Search: Finds closest vectors/items.
Cosine Similarity: Compares vector orientation.
Euclidean Distance: Measures straight-line distance.
Clustering Analysis: Groups similar records.
K-means: Clusters around centroids.
Hierarchical Clustering: Builds nested cluster tree.
Spectral Clustering: Uses graph structure for clustering.
Graph Analytics: Analyzes nodes and edges.
Knowledge Graph: Structured entity-relation graph.

Processes

Data Collection: Gather raw data.
Data Cleaning: Fix missing, duplicate, inconsistent data.
Deduplication: Removes repeated records.
Filtering: Removes low-quality or irrelevant data.
Data Transformation: Convert data into model-ready form.
Feature Scaling: Normalize numeric ranges.
Categorical Encoding: Convert categories to numbers.
Tokenization: Break text into tokens.
Chunking: Split documents for retrieval/context limits.
Data Labeling: Assign ground-truth labels.
Data Versioning: Track dataset changes.
Dataset Provenance: Track data origin and usage rights.
Train/Validation/Test Split: Separate training, tuning, final testing.
Insight Generation: Turn analysis into action.

Tools

RAPIDS: GPU-accelerated data science suite.
cuDF: GPU DataFrame library; pandas-like.
cuML: GPU ML algorithms.
cuGraph: GPU graph analytics.

Dask-cuDF: Distributed multi-GPU DataFrames.

NVTabular: GPU feature engineering for tabular data.

CUDA-X: NVIDIA accelerated software libraries.

NVIDIA DALI: GPU data loading/preprocessing.

TAO Toolkit: Simplifies training and augmentation.

Pandas: CPU data manipulation.

Polars: Fast DataFrame library.

Dask: Distributed Python data processing.

Experimentation & Evaluation

Weight: 22%

Methods

A/B Testing: Compare versions on real traffic.

Benchmarking: Compare against standard baselines.

Cross-validation: Evaluate across multiple data splits.

K-fold Validation: Split data into k train/test folds.

Hold-out Testing: Use separate unseen test set.

Temporal Validation: Test using future time data.

Statistical Testing: Checks significance of results.

Human Evaluation: Humans judge quality.

LLM-as-a-Judge: LLM scores outputs.

Error Analysis: Systematically inspect failures.

Ablation Study: Remove components to measure impact.

Stress Testing: Test under heavy/extreme conditions.

Adversarial Testing: Test with hostile edge cases.

Red Teaming: Attack model to find risks.

Canary Deployment: Release to small user subset.

Metrics

Accuracy: Fraction correct.

Precision: Correct positives among predicted positives.

Recall: Correct positives among actual positives.

F1 Score: Harmonic mean of precision and recall.

Confusion Matrix: Counts prediction categories.

Specificity: Correct negatives among actual negatives.

AUC-ROC: Ranking quality across thresholds.

MAE: Average absolute regression error.

MSE: Average squared regression error.
RMSE: Square root of MSE.
R-squared: Regression variance explained.
Cross-entropy Loss: Classification probability loss.
BLEU: N-gram overlap for translation.
ROUGE: Overlap metric for summaries.
METEOR: Translation quality metric.
Perplexity: Language model uncertainty.
Groundedness: Output supported by sources.
Faithfulness: Response matches retrieved evidence.
Relevance: Answer addresses the query.
Coherence: Output is logically organized.
Toxicity Score: Harmful language measurement.
Latency: Time per request.
Throughput: Requests/tokens per time.
Error Rate: Failed requests percentage.
GPU Utilization: How much GPU is used.

RLHF

RLHF: Aligns LLM using human feedback.
Human Feedback: Human preference labels.
Preference Pair: Two outputs ranked by humans.
Reward Model: Predicts human preference score.
PPO: RL algorithm used in RLHF.
DPO: Directly optimizes preference data.
Reward Hacking: Model exploits reward incorrectly.
Safety Alignment: Makes outputs follow human values.
Constitutional AI: Uses written principles for alignment.
SFT: Supervised fine-tuning on instruction-response pairs.

Software Development

Weight: 24%

Development

Model Deployment: Serve models in production.
API Endpoint: Network interface for model calls.
REST API: HTTP-based service interface.
gRPC: High-performance service protocol.

SDK: Developer tools for using a platform.

Containerization: Package app and dependencies.

Docker: Container runtime.

Kubernetes: Container orchestration system.

Helm Chart: Kubernetes deployment package.

Version Control: Track code/model changes.

Git: Distributed version control tool.

CI/CD: Automated build, test, deployment.

Unit Testing: Tests small code units.

Integration Testing: Tests connected components.

Model Validation Pipeline: Automated model quality checks.

Configuration File: Stores settings/hyperparameters.

YAML/JSON Config: Common configuration formats.

Logging: Records events for debugging/audit.

Asynchronous Programming: Handles many I/O tasks concurrently.

Scalability: Handles growing workload.

Horizontal Scaling: Add more instances.

Vertical Scaling: Add more resources per instance.

Caching: Reuse frequent responses/results.

Frameworks and App Tools

LangChain: Framework for LLM apps.

LangGraph: Orchestrates complex LLM workflows.

ChatNVIDIA: LangChain connector for NVIDIA chat models.

NVIDIAEmbeddings: LangChain connector for NVIDIA embeddings.

Vector Database: Stores/searches embeddings.

FAISS: Vector similarity search library.

Milvus: Vector database.

Pinecone: Managed vector database.

Weaviate: Vector database with semantic search.

ONNX: Open model exchange format.

OpenAI-compatible API: Standard chat/completion API shape.

Monitoring

Performance Monitoring: Tracks latency/throughput/errors.

Model Monitoring: Tracks quality/drift.

Infrastructure Monitoring: Tracks CPU/GPU/memory.

Drift Detection: Finds distribution changes.
Quality Degradation Alert: Warns when model worsens.
Automated Alerts: Real-time issue notifications.
Audit Trail: Record of actions and outputs.
User Feedback Loop: Uses real usage feedback.

Trustworthy AI Principles

Weight: 10%

Ethics

Fairness: Equitable treatment across groups.
Transparency: Decisions are understandable.
Accountability: Clear responsibility for outcomes.
Privacy: Protect sensitive data.
Consent: Respect user data permission.
Beneficence: System should benefit people.
Robustness: Performs reliably under variation.
Explainability: Explains model decisions.
Compliance: Follows laws and regulations.
Model Card: Documents model purpose, limits, risks.
Data Sheet: Documents dataset details.

Bias

AI Bias: Unfair systematic model behavior.
Data Bias: Bias from training data.
Sampling Bias: Unrepresentative data collection.
Label Bias: Biased annotation/ground truth.
Algorithmic Bias: Bias introduced by model/design.
Evaluation Bias: Narrow tests hide failures.
Bias Detection: Measure unfair group differences.
Bias Mitigation: Reduce unfair outcomes.
Fairness Constraints: Enforce fairness during optimization.
Adversarial Debiasing: Train model to reduce bias signals.
Diverse Test Sets: Evaluate across groups/scenarios.

Safety and Guardrails

NeMo Guardrails: Controls safe LLM behavior.
Colang: Language for Guardrails dialogue flows.

Input Rail: Filters user input before LLM.
Output Rail: Filters response after LLM.
Topical Rail: Keeps conversation on allowed topics.
Fact-checking Rail: Verifies claims against sources.
Jailbreak: Prompt that bypasses safety rules.
Prompt Injection: Malicious instruction inside input/context.
Toxicity Detection: Finds harmful language.
PII Protection: Protects personal information.
Differential Privacy: Adds privacy-preserving noise.

Key NVIDIA Technologies

Platforms

NVIDIA NIM: GPU-optimized inference microservice containers.
NVIDIA NeMo: Framework for LLM training/fine-tuning/customization.
TensorRT-LLM: Optimizes LLM inference on NVIDIA GPUs.
Triton Inference Server: Serves models with batching/versioning.
RAPIDS: GPU-accelerated data science libraries.
cuDF: GPU DataFrame library.
cuML: GPU ML algorithms.
cuGraph: GPU graph analytics.
CUDA: NVIDIA GPU programming platform.
cuDNN: GPU primitives for deep neural networks.
NCCL: Multi-GPU communication library.
NGC: NVIDIA catalog for models, containers, SDKs.
BioNeMo: Foundation models for biology/life sciences.
Megatron-LM: Large-scale transformer training framework.
NeMo Curator: Data curation/preprocessing pipeline.
NeMo Guardrails: Safety framework for LLM apps.
NVIDIA AI Endpoints: Hosted NVIDIA model APIs.
DeepStream SDK: Real-time video AI streaming.
Omniverse: 3D simulation/visualization platform.
Omniverse Nucleus: Collaboration/storage for Omniverse assets.

NIM Details

NIM Container: Prebuilt optimized model server.
NGC Distribution: NIM images come from NGC.
OpenAI API Compatibility: Existing clients can switch easily.

REST/gRPC Endpoint: NIM exposes production APIs.
TensorRT-LLM inside NIM: Provides optimized inference.
Triton Backend: Many NIMs use Triton internally.
Hardware Portability: Same container across cloud/on-prem/edge.

NeMo Details

SFT in NeMo: Trains on prompt-response pairs.
RLHF in NeMo: Reward model plus PPO alignment.
LoRA in NeMo: Efficient adapter fine-tuning.
P-Tuning: Learns soft prompt vectors.
Prompt Tuning: Trains prompt embeddings.
Distributed Training: Scales across GPUs/nodes.
Custom Tokenizer: Domain-specific token splitting.
Megatron Integration: Enables tensor/pipeline parallelism.

TensorRT-LLM Details

In-flight Batching: Adds requests into active batches.
Continuous Batching: Keeps GPU busy during serving.
Paged KV Cache: Pages attention cache memory.
Custom CUDA Kernels: Faster low-level GPU operations.
Quantization: Lower precision for speed/memory.
INT8: 8-bit inference precision.
INT4: 4-bit compression precision.
FP8: Low precision floating-point.
AWQ: Activation-aware weight quantization.
GPTQ: Post-training quantization method.
Tensor Parallelism: Splits tensors across GPUs.

Triton Details

Dynamic Batching: Groups requests automatically.
Model Ensemble: Chains models into one pipeline.
Model Versioning: Hosts multiple versions.
Concurrent Execution: Runs multiple models together.
Multi-framework Serving: Supports PyTorch, TF, ONNX, TensorRT.
Model Repository: Directory of deployable models/configs.
A/B Testing: Compare model versions in production.

NGC Details

Model Catalog: Find pretrained models.
NIM Catalog: Get production containers.
Helm Charts: Deploy on Kubernetes.
Version Pinning: Reproducible container/model versions.
Model Signing: Verifies container integrity.
Security Scanning: Checks container vulnerabilities.

Prompt Engineering Mastery

Techniques

Prompt Engineering: Craft prompts for desired outputs.
Zero-shot Prompting: No examples provided.
Few-shot Prompting: Includes examples in prompt.
One-shot Prompting: Includes one example.
Chain-of-Thought: Encourages step-by-step reasoning.
Self-consistency: Sample multiple reason paths; choose consensus.
Tree of Thoughts: Explore multiple reasoning branches.
Role Prompting: Assign model a role/persona.
System Prompt: Sets top-level behavior/instructions.
Delimiters: Separate instructions from data.
Structured Output Prompting: Request JSON/tables/schema.
Output Parsing: Convert model output to usable format.
Prompt Chaining: Use outputs as later inputs.
Function Calling: Let LLM call external tools.
Multi-turn Conversation: Maintain dialogue context.
Domain Adaptation Prompting: Tailor prompt to domain.
Safety Prompting: Reduce harmful/off-policy responses.

Generation Controls

Temperature: Controls randomness.
Low Temperature: More deterministic output.
High Temperature: More creative output.
Top-k Sampling: Sample from k most likely tokens.
Top-p Sampling: Sample from probability nucleus.
Max Tokens: Limits output length.
Frequency Penalty: Discourages repeated tokens.
Repetition Penalty: Reduces repeated phrases.
Context Management: Use context window efficiently.

Token Efficiency: Minimize tokens while preserving task.

Iterative Refinement: Improve prompt based on output.

Model Evaluation & Testing

Evaluation

Automated Metrics: Quantitative model scores.

Human Evaluation: Expert/user quality judgment.

Benchmark Dataset: Standard comparison dataset.

GLUE: NLP understanding benchmark.

SuperGLUE: Harder NLP benchmark.

HELM: Holistic LLM evaluation benchmark.

MMLU: Multi-domain knowledge benchmark.

Domain-specific Test: Custom test for target use case.

Adversarial Test: Robustness against attacks/edge cases.

Regression Test: Ensures behavior does not break.

Golden Dataset: Trusted reference test set.

Validation

Cross-validation: Reliable performance estimate.

Hold-out Test: Final unbiased test set.

Temporal Validation: Future data validation.

Distribution Shift: Train/test data differ.

Stress Test: Performance under heavy load.

Calibration: Confidence matches correctness.

Generalization: Works on unseen examples.

Monitoring

Drift Detection: Input/output distribution changes.

Performance Tracking: Continuous metric monitoring.

Error Analysis: Investigate failure patterns.

User Feedback: Real-world quality signal.

Automated Alert: Notifies degradation.

Production Evaluation: Measures live system behavior.

Repeated Exam Patterns

Tool-to-Phase Mapping

Data Prep: RAPIDS, cuDF, DALI, NeMo Curator.

Training: NeMo, Megatron-LM, PyTorch, TensorFlow.
Fine-tuning: NeMo, SFT, LoRA, P-Tuning, Prompt Tuning.
Alignment: RLHF, reward model, PPO, DPO.
Optimization: TensorRT-LLM, quantization, batching, KV cache.
Serving: NIM, Triton, REST/gRPC.
App Development: LangChain, LangGraph, NVIDIA AI Endpoints.
Retrieval: Embeddings, vector DB, FAISS, Milvus.
Safety: NeMo Guardrails, Colang, rails.
Catalog: NGC.
Monitoring: Logs, drift, latency, throughput, alerts.

RAG Pipeline

User Query: User asks a question.
Query Embedding: Convert query to vector.
Vector Database: Stores searchable document vectors.
ANN Search: Fast approximate nearest neighbor retrieval.
Retriever: Finds relevant chunks.
Re-ranker: Reorders retrieved chunks by relevance.
Context Injection: Add chunks into prompt.
LLM Generation: Produce grounded answer.
Citation/Source Check: Verify answer support.
RAG Benefit: Reduces hallucination and adds fresh knowledge.
RAG Risk: Bad retrieval causes bad answers.

Fine-tuning Methods

Full Fine-tuning: Updates all model weights.
PEFT: Updates small trainable components.
LoRA: Low-rank trainable adapters.
QLoRA: Quantized base model plus LoRA.
Adapter Tuning: Small modules inserted in layers.
Prompt Tuning: Trains soft prompt tokens.
P-Tuning: Learns continuous prompt vectors.
Instruction Tuning: Teaches instruction following.
Domain Adaptation: Adapts model to specialized domain.
Catastrophic Forgetting: New training erases old skills.
Gradient Checkpointing: Saves memory by recomputing activations.
Mixed Precision Training: Uses FP16/BF16 for speed/memory.

Optimization and Deployment

Quantization: Reduces precision for efficient inference.
Pruning: Removes less important weights.
Distillation: Smaller model learns from larger model.
Batching: Process requests together.
Dynamic Batching: Triton groups concurrent requests.
Continuous/In-flight Batching: Adds requests mid-generation.
Speculative Decoding: Smaller draft model speeds generation.
KV Cache: Speeds autoregressive decoding.
Caching: Stores common responses/results.
Autoscaling: Adjusts replicas to demand.
Cold Start: Delay when starting service/model.
Edge Deployment: Run model near device/user.
Model Registry: Tracks model versions/artifacts.
Canary Release: Small rollout before full release.
Container Orchestration: Manage containers at scale.

Trust and Risk

Hallucination: Confident but false output.
Grounding: Connect output to evidence.
Jailbreak: Bypass safety guardrails.
Prompt Injection: Malicious prompt hidden in input.
Data Leakage: Sensitive data exposed.
Copyright Risk: Training data may violate rights.
Privacy Risk: User data mishandled.
Bias Risk: Unequal performance across groups.
Toxic Output: Harmful or offensive response.
Robustness Risk: Fails under unusual inputs.
Accountability Gap: Unclear responsibility for harm.

Data Types and Preprocessing

Image Data: Resize, normalize, augment.
Text Data: Tokenize, pad, embed.
Audio Data: Spectrograms, normalization, augmentation.
Tabular Data: Scale numbers, encode categories.
Time-series Data: Normalize, window, forecast.
Image Rotation: Augments image angle.

Image Flipping: Mirrors images.
Random Crop: Varies image view.
Color Jitter: Changes brightness/contrast/color.
Noise Injection: Adds random noise for robustness.
Synonym Replacement: Text augmentation by synonyms.
Back Translation: Translate out and back to paraphrase.
Random Insertion/Deletion: Adds/removes words.
Time Shift: Audio time augmentation.
Pitch Scaling: Audio pitch augmentation.

Common Confusions

NIM vs Triton: NIM is packaged model microservice; Triton is serving server.
NIM vs NGC: NIM runs models; NGC stores/distributes containers/models.
NeMo vs NIM: NeMo trains/customizes; NIM deploys/serves.
TensorRT-LLM vs Triton: TensorRT-LLM optimizes LLMs; Triton serves models.
Guardrails vs RLHF: Guardrails control runtime behavior; RLHF changes model alignment.
RAPIDS vs Pandas: RAPIDS/cuDF runs DataFrames on GPU; pandas runs on CPU.
RAG vs Fine-tuning: RAG retrieves knowledge; fine-tuning changes weights.
LoRA vs Full Fine-tuning: LoRA updates adapters; full updates all weights.
BLEU vs ROUGE: BLEU often translation; ROUGE often summarization.
Precision vs Recall: Precision avoids false positives; recall avoids false negatives.
Latency vs Throughput: Latency is delay; throughput is volume.
Encoder vs Decoder: Encoder understands; decoder generates.
Self-attention vs Cross-attention: Same sequence vs encoder-decoder attention.

AI Subject Terms Glossary

Core AI Terms

Agent: System that acts toward a goal.
AI Pipeline: Steps from data to deployed model.
Algorithm: Procedure for solving a task.
Baseline: Simple reference model/performance.
Checkpoint: Saved model state.
Class Imbalance: Unequal class distribution.
Data Leakage: Test information enters training.
Decision Boundary: Separates predicted classes.
Embedding Space: Vector space of learned meanings.
Feature: Input variable used by model.

Generalization: Performs well on unseen data.
Ground Truth: Correct reference answer/label.
Heuristic: Rule-of-thumb method.
Label: Correct output for supervised training.
Latent Space: Compressed hidden representation.
Model Architecture: Structure of model layers/components.
Objective Function: Function optimized during training.
Pretraining: Training on broad large data first.
Prompt: Input instructions sent to LLM.
Representation: Learned encoding of data.
Sampling: Choosing output tokens probabilistically.
Training Corpus: Dataset used for training.
Validation Set: Data used for tuning decisions.

Neural Network Terms

Activation Function: Adds nonlinearity to networks.
ReLU: Outputs $\max(0, x)$.
Sigmoid: Maps values to 0-1.
Tanh: Maps values to -1 to 1.
GELU: Smooth activation common in transformers.
Dead ReLU: ReLU stuck outputting zero.
Gradient: Direction/size of parameter update.
Vanishing Gradient: Gradients become too small.
Exploding Gradient: Gradients become too large.
Optimizer: Updates model weights.
AdamW: Adaptive optimizer with weight decay.
SGD: Basic gradient descent optimizer.
Weight Decay: Regularization penalizing large weights.
Batch Normalization: Normalizes layer inputs.
Layer Normalization: Normalizes features per sample.
Residual Block: Layer with skip connection.
Parameter Count: Number of learned weights.
Compute: Processing needed for training/inference.
FLOPs: Floating-point operations.
VRAM: GPU memory.

NLP and LLM Terms

NLP: AI for human language.
Token: Basic text unit processed by model.
Tokenizer: Converts text into tokens.
Vocabulary: Set of tokenizer tokens.
Subword Tokenization: Splits words into pieces.
BPE: Byte Pair Encoding tokenizer method.
WordPiece: Subword tokenizer used by BERT.
SentencePiece: Language-independent tokenizer.
Padding Token: Filler token for equal lengths.
Attention Mask: Marks real tokens vs padding.
Autoregressive Model: Predicts next token from previous tokens.
Masked Language Modeling: Predicts hidden tokens.
Next Sentence Prediction: Predicts sentence continuity.
Causal Language Modeling: Next-token prediction.
Bidirectional Attention: Looks left and right.
Unidirectional Attention: Looks only previous tokens.
Instruction Following: Model obeys user task instructions.
In-context Learning: Learns from prompt examples.
Context Length: Maximum tokens accepted.
Stop Sequence: Text that ends generation.
Logit: Raw score before probability.
Softmax: Converts scores to probabilities.
Decoding: Converts model probabilities to text.
Greedy Decoding: Always picks most likely token.
Beam Search: Keeps multiple candidate sequences.
Nucleus Sampling: Top-p sampling.

Generative AI Terms

Generative Model: Creates new data.
Discriminative Model: Predicts labels/classes.
Generator: GAN network that creates samples.
Discriminator: GAN network that judges samples.
Forward Diffusion: Adds noise to data.
Reverse Diffusion: Removes noise to generate.
Denoising: Removing noise from data.
Latent Diffusion: Diffusion in compressed space.
Multimodal Model: Handles multiple input types.

Image Captioning: Generates text description of image.
Text-to-Image: Generates image from prompt.
Image-to-Text: Generates text from image.
Speech-to-Text: Converts audio to text.
Text-to-Speech: Converts text to audio.
Synthetic Data: Artificially generated training data.
Hallucination: Plausible but incorrect generation.
Grounded Generation: Generation supported by evidence.

Evaluation Terms

Benchmark: Standard test for comparison.
Metric: Numeric performance measure.
Loss: Training error signal.
Train Loss: Error on training data.
Validation Loss: Error on validation data.
Test Accuracy: Final held-out accuracy.
False Positive: Incorrect positive prediction.
False Negative: Incorrect negative prediction.
True Positive: Correct positive prediction.
True Negative: Correct negative prediction.
ROC Curve: TPR/FPR curve across thresholds.
Confidence Score: Model certainty estimate.
Calibration Error: Confidence mismatch.
Inter-rater Agreement: Human judge consistency.
Regression Metric: Score for numeric prediction.
Classification Metric: Score for class prediction.
Retrieval Metric: Score for search quality.
Recall@k: Relevant item appears in top k.
MRR: Mean reciprocal rank.
NDCG: Ranking quality metric.

Data Science Terms

Grid Search: Tests all parameter combinations.
Randomized Search: Samples parameter combinations.
Hyperparameter Tuning: Finds best config.
Cross-validation Fold: One split in k-fold testing.
ARIMA: Time-series forecasting model.

Autocorrelation: Correlation with past values.
Partial Autocorrelation: Direct lag correlation.
Stationarity: Stable mean/variance over time.
Seasonality: Repeating time pattern.
Trend: Long-term direction.
Intercept: Baseline regression value.
Coefficient: Feature effect in regression.
Residual: Prediction error.
Outlier: Unusual data point.
Missing Value: Empty/unknown feature value.
Imputation: Fill missing values.
One-hot Encoding: Converts categories to binary columns.

NVIDIA and Production Terms

MIG: Splits one GPU into isolated instances.
Time-sharing: Multiple workloads share GPU time.
Model Signing: Cryptographic verification of model/container.
Vulnerability Scanning: Checks security weaknesses.
Version Pinning: Locks exact model/container version.
Helm: Kubernetes package manager.
Load Balancing: Distributes traffic across servers.
SLA: Required service reliability/performance.
Observability: Logs, metrics, traces for systems.
Telemetry: Collected runtime measurements.
Throughput Bottleneck: Component limiting system speed.
CPU-GPU Transfer: Data movement between CPU and GPU.
Zero-copy Interop: Avoids unnecessary data copying.

Python Tools and Libraries

Core Python ML Stack

Python: Main language for AI/ML workflows.
NumPy: Fast numerical arrays and math.
Pandas: CPU DataFrame data analysis.
Polars: Fast DataFrame processing library.
SciPy: Scientific computing and optimization.
scikit-learn: Classical ML and metrics toolkit.
Jupyter Notebook: Interactive coding and exploration.

Matplotlib: Basic Python plotting.

Seaborn: Statistical visualization on Matplotlib.

Plotly: Interactive charts and dashboards.

Statsmodels: Statistical models and time-series tools.

NLTK: Traditional NLP toolkit.

spaCy: Production NLP pipeline library.

Deep Learning Libraries

PyTorch: Flexible deep learning framework.

torch: PyTorch tensor and neural network package.

torch.nn: PyTorch neural network layers.

torch.optim: PyTorch optimizers.

torchvision: PyTorch image models/transforms.

torchaudio: PyTorch audio processing.

TensorFlow: Production deep learning framework.

Keras: High-level TensorFlow model API.

TensorBoard: Training visualization dashboard.

JAX: High-performance array/autodiff framework.

Autograd: Automatic gradient computation.

CUDA Tensor: Tensor stored on NVIDIA GPU.

Hugging Face Ecosystem

Transformers: Pretrained transformer models library.

AutoModel: Loads model architecture automatically.

AutoTokenizer: Loads matching tokenizer automatically.

Datasets: Dataset loading and preprocessing library.

Tokenizers: Fast tokenizer implementations.

Accelerate: Simplifies distributed/mixed-precision training.

PEFT: Parameter-efficient fine-tuning library.

TRL: RLHF/DPO training utilities.

Safetensors: Safe fast model weight format.

Model Hub: Repository for pretrained models/datasets.

Pipeline API: Quick task-based inference wrapper.

LLM App Development

LangChain: Builds LLM apps and chains.

LangGraph: Stateful graph workflows for agents.

LlamaIndex: Data connectors and RAG framework.

OpenAI Python SDK: Calls OpenAI-compatible APIs.
NVIDIA AI Endpoints: Hosted NVIDIA model APIs.
ChatNVIDIA: LangChain chat connector for NVIDIA.
NVIDIAEmbeddings: LangChain embedding connector.
FAISS Python: Local vector similarity search.
Milvus SDK: Connects Python to Milvus vector DB.
Pinecone SDK: Connects Python to Pinecone.
Weaviate Client: Connects Python to Weaviate.
Chroma: Lightweight local vector database.
SentenceTransformers: Embedding models for semantic search.
tiktoken: Token counting/tokenization utility.

NVIDIA Python and GPU Tools

RAPIDS: GPU data science Python ecosystem.
cuDF: GPU DataFrame library.
cudf.pandas: GPU acceleration for pandas code.
cuML: GPU machine learning algorithms.
cuGraph: GPU graph analytics.
Dask-cuDF: Distributed GPU DataFrames.
CuPy: NumPy-like arrays on GPU.
Numba: JIT compiler for Python/GPU kernels.
NVIDIA DALI: GPU data loading and preprocessing.
NVTabular: GPU tabular feature engineering.
NeMo Toolkit: Python framework for LLM customization.
NeMo Curator: Dataset curation and filtering.
NeMo Guardrails: Python safety/rail framework.
TensorRT Python API: Build optimized inference engines.
TensorRT-LLM: Optimized LLM inference library.
Triton Client: Python client for Triton inference.
NVIDIA NIM Client Pattern: Call NIM via REST/gRPC.
NCCL Bindings: Multi-GPU communication support.

Data Processing Libraries

PyArrow: Columnar data and Parquet support.
Parquet: Efficient columnar storage format.
JSONL: One JSON object per line.
CSV: Simple tabular text format.

BeautifulSoup: HTML parsing.

Requests: HTTP API calls.

aiohttp: Async HTTP client/server.

Regex: Pattern matching for text cleaning.

Unidecode: Text normalization utility.

pydantic: Data validation and schemas.

Serving and API Libraries

FastAPI: Modern Python REST API framework.

Flask: Lightweight Python web framework.

Uvicorn: ASGI server for FastAPI.

Gunicorn: Production Python web server.

gRPC Python: High-performance RPC services.

Streamlit: Quick Python data apps.

Gradio: Quick ML demos/interfaces.

ONNX Runtime: Runs ONNX models.

Triton Inference Server: Production model serving.

Docker SDK: Control Docker from Python.

Kubernetes Client: Manage Kubernetes from Python.

Testing and Quality Tools

pytest: Python testing framework.

unittest: Built-in Python test framework.

mypy: Static type checking.

ruff: Fast Python linter/formatter.

black: Opinionated Python formatter.

pre-commit: Runs checks before commits.

tox: Tests across environments.

coverage.py: Measures test coverage.

Great Expectations: Data quality validation.

Evidently: ML drift and monitoring reports.

Experiment Tracking and MLOps

MLflow: Tracks experiments and models.

Weights & Biases: Experiment tracking dashboard.

DVC: Data/model version control.

Hydra: Configuration management.

OmegaConf: Hierarchical config library.

Airflow: Workflow orchestration.

Prefect: Python workflow orchestration.

Ray: Distributed Python computing.

Ray Serve: Scalable model serving.

BentoML: Package and serve ML models.

Python Concepts for Certification

Virtual Environment: Isolates project dependencies.

pip: Python package installer.

conda: Environment and package manager.

requirements.txt: Lists pip dependencies.

pyproject.toml: Modern Python project config.

Asyncio: Python async concurrency.

Coroutine: Async function execution unit.

Type Hint: Optional Python type annotation.

Exception Handling: Handles runtime errors.

Serialization: Save/load data structures.

Pickle: Python object serialization.

Environment Variable: Runtime configuration value.

Logging Module: Built-in structured logging.

Exam Decision Tree

NVIDIA Tool Choice

Need deploy model fast: Choose NVIDIA NIM.

Need OpenAI-compatible endpoint: Choose NVIDIA NIM.

Need private/on-prem inference: Choose NVIDIA NIM.

Need optimize LLM inference: Choose TensorRT-LLM.

Need lower latency: Choose TensorRT-LLM or batching.

Need higher throughput: Choose TensorRT-LLM, Triton, batching.

Need serve many frameworks: Choose Triton.

Need model versioning/A-B tests: Choose Triton.

Need chain models server-side: Choose Triton ensemble.

Need train or fine-tune LLM: Choose NeMo.

Need RLHF/SFT/LoRA: Choose NeMo.

Need curate training data: Choose NeMo Curator.

Need GPU DataFrame processing: Choose RAPIDS/cuDF.

Need GPU ML algorithms: Choose cuML.

Need GPU graph analytics: Choose cuGraph.
Need safety/policy controls: Choose NeMo Guardrails.
Need block jailbreaks/injections: Choose Guardrails rails.
Need find NVIDIA containers/models: Choose NGC.
Need biology foundation models: Choose BioNeMo.
Need video AI streaming: Choose DeepStream.
Need GPU preprocessing: Choose DALI.

Model Method Choice

Need external/current knowledge: Use RAG.
Need domain behavior/style: Use fine-tuning.
Need cheap fine-tuning: Use PEFT/LoRA.
Need align to preferences: Use RLHF/DPO.
Need reduce model size: Use distillation/pruning/quantization.
Need faster decoding: Use KV cache/speculative decoding.
Need reduce memory: Use quantization/LoRA/checkpointing.
Need compare live versions: Use A/B testing.
Need unbiased final score: Use held-out test set.
Need robust estimate: Use cross-validation.

Common Exam Traps

RAG vs Fine-tuning: RAG adds knowledge; fine-tuning changes behavior/weights.
NIM vs NGC: NIM runs models; NGC stores models/containers.
NIM vs Triton: NIM is packaged service; Triton is serving infrastructure.
NeMo vs NIM: NeMo trains/customizes; NIM deploys.
TensorRT-LLM vs Triton: TensorRT optimizes; Triton serves.
Guardrails vs RLHF: Guardrails runtime controls; RLHF training alignment.
cuDF vs Pandas: cuDF GPU; pandas CPU.
cuML vs scikit-learn: cuML GPU ML; scikit-learn CPU ML.
Latency vs Throughput: Latency delay; throughput volume.
Precision vs Recall: Precision reduces false positives; recall reduces false negatives.
F1 vs Accuracy: F1 better for imbalance; accuracy can mislead.
BLEU vs ROUGE: BLEU translation; ROUGE summarization.
Perplexity vs Accuracy: Perplexity language uncertainty; accuracy exact correctness.
Encoder-only vs Decoder-only: Encoder understands; decoder generates.
BERT vs GPT: BERT bidirectional encoder; GPT autoregressive decoder.
Self-attention vs Cross-attention: Same sequence vs encoder-decoder link.

Top-k vs Top-p: Fixed count vs probability mass.
Temperature 0 vs High: Deterministic vs creative.
Quantization vs Pruning: Lower precision vs remove weights.
Data Parallelism vs Tensor Parallelism: Split data vs split model tensors.
Pipeline Parallelism vs Tensor Parallelism: Split layers vs split tensor operations.
Validation Set vs Test Set: Tune on validation; final score on test.
Data Bias vs Algorithmic Bias: Bad data vs biased model/design.
Prompt Injection vs Jailbreak: Hidden malicious instruction vs bypass attempt.
Groundedness vs Relevance: Supported by sources vs answers question.

Formula and Metric Mini Sheet

Classification

Accuracy: Correct predictions / all predictions.
Precision: $TP / (TP + FP)$.
Recall: $TP / (TP + FN)$.
F1: $2 * \text{precision} * \text{recall} / (\text{precision} + \text{recall})$.
False Positive: Predicted positive, actually negative.
False Negative: Predicted negative, actually positive.
Confusion Matrix: Table of TP, FP, FN, TN.
AUC-ROC: Ranking quality across thresholds.

Regression

MAE: Average absolute error.
MSE: Average squared error.
RMSE: Square root of MSE.
R-squared: Proportion of variance explained.

NLP and LLM

Perplexity: Lower means better next-token prediction.
Cross-entropy: Penalizes wrong probability predictions.
BLEU: N-gram precision for translation.
ROUGE: N-gram recall for summarization.
METEOR: Translation metric with synonym/stem matching.
Groundedness: Claims supported by context.
Faithfulness: Output does not contradict source.
Toxicity: Harmful/offensive content score.

Retrieval and RAG

Cosine Similarity: Measures vector angle similarity.
Recall@k: Relevant item appears in top k.
Precision@k: Relevant fraction in top k.
MRR: Average reciprocal rank of first relevant result.
NDCG: Ranking quality with position weighting.

Production

Latency: Time for one request.
Throughput: Requests/tokens per second.
Error Rate: Failed requests / total requests.
GPU Utilization: Percent GPU busy.
Memory Footprint: RAM/VRAM required.
Tokens/sec: Generation speed.

Scenario Keywords

NVIDIA Tools

"OpenAI-compatible API": NIM.
"Pre-packaged optimized container": NIM.
"Run model locally/private cloud": NIM.
"Pull from nvcr.io": NGC/NIM.
"Catalog of models/containers": NGC.
"Helm chart": NGC/Kubernetes deployment.
"Fine-tune LLM": NeMo.
"SFT/RLHF/LoRA/P-Tuning": NeMo.
"Reward model/PPO": RLHF with NeMo.
"Data curation/filtering": NeMo Curator.
"Block unsafe topics": NeMo Guardrails.
"Colang": NeMo Guardrails.
"Input/output rails": NeMo Guardrails.
"Optimize LLM latency": TensorRT-LLM.
"Paged KV cache": TensorRT-LLM.
"INT4/INT8/FP8": TensorRT-LLM quantization.
"Dynamic batching": Triton.
"Model ensemble": Triton.
"Model versioning": Triton.

"Serve PyTorch/TensorFlow/ONNX together": Triton.

"GPU pandas": cuDF.

"GPU data science": RAPIDS.

"GPU clustering/PCA/nearest neighbors": cuML.

"GPU graph analytics": cuGraph.

Model and Data Methods

"Needs latest/private documents": RAG.

"Reduce hallucinations with sources": RAG.

"Vector database": Embeddings/RAG.

"Semantic search": Embeddings/vector DB.

"Small GPU memory for tuning": LoRA/QLoRA.

"Train only few parameters": PEFT.

"Human preferences": RLHF/DPO.

"Pairwise comparisons": Reward model/DPO.

"Reduce model size": Distillation/pruning/quantization.

"Faster inference": Quantization, TensorRT-LLM, batching.

"Multiple GPUs too large model": Tensor parallelism.

"Split layers across GPUs": Pipeline parallelism.

"Same model copy on each GPU": Data parallelism.

Evaluation and Safety

"Imbalanced classes": F1/precision/recall.

"False positives costly": Precision.

"False negatives costly": Recall.

"Translation quality": BLEU/METEOR.

"Summarization quality": ROUGE/human eval.

"Language model fluency": Perplexity.

"Open-ended answers": Human eval/LLM-as-judge.

"Production comparison": A/B testing.

"Final unbiased score": Hold-out test set.

"Distribution changes": Drift detection.

"Harmful prompts": Red teaming/adversarial testing.

"Sensitive user data": Privacy/PII controls.

"Demographic fairness": Bias/fairness evaluation.